

ASTRA INTEL | Unclassified // Training sample
Unmanned Aerial Systems Capability Brief 2024

1. Purpose

This brief outlines current unmanned aerial systems (UAS) roles in intelligence, surveillance and reconnaissance (ISR) and tactical strike.

Group 2-3 UAS platforms provide persistent ISR coverage over contested littoral approaches without exposing aircrew to surface-to-air threats.

2. Force design

A mixed fleet of medium-altitude long-endurance air vehicles and smaller tactical quadcopters is recommended for brigade and joint task force use.

Attritable airframes should be stocked to absorb expected combat loss rates.

3. Command and control

Beyond visual line of sight operations require redundant C2 links, including SATCOM primary and line-of-sight radio as a fallback path. Ground control stations should implement STANAG 4586 interoperability so allied payloads and air vehicles can be retasked during coalition ops.

Lost-link procedures must include a documented return-to-home orbit and an automatic flight termination option over water or designated ranges.

4. Sensors and payloads

Electro-optical / infrared turrets remain the primary ISR payload.

Synthetic aperture radar is required for weather-independent mapping of coastal movement corridors in the Indo-Pacific theatre.

Electronic support measures can cue other sensors onto emitters of interest.

Strike variants may carry precision-guided munitions only under positive human authorisation. Autonomous target engagement is not authorised.

5. Major applications

Border and exclusive economic zone monitoring.

Convoy overwatch and route reconnaissance for land manoeuvre units.

Ship-to-shore surveillance during amphibious entry operations.

Battle damage assessment after stand-off fires.

6. Risks and limitations

UAS are vulnerable to cheap electronic warfare and GNSS spoofing.

High-bandwidth video is difficult to sustain under contested spectrum.

Public airspace integration still constrains training sorties near cities.

Weather, icing and icing-adjacent humidity reduce small-UAS endurance.

7. Key organisations and systems

MQ-9 Reaper and similar MALE types remain the theatre ISR workhorse. General Atomics Aeronautical Systems and allied industry partners supply airframes, while NATO working groups set payload interface standards. National geospatial agencies consume the resulting imagery products.

Dates of interest: capability refresh programmed for 2024-2026, with a follow-on attritable system decision expected in late 2026.

8. Summary

Persistent ISR from Group 2-3 UAS is a force multiplier if C2 resilience, spectrum management and airspace integration are funded together.

The recommended near-term focus is redundant datalinks, SAR payloads, and training for operations in GNSS-degraded environments.